# Roll No:CO-2105 and Name:Sandhya Raghunath Magadum

In [ ]:

## 1. Check whether a number is even or odd

In [ ]:
```
num = int(input("Enter a number: "))
if num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

## 2. Check whether a number is positive, negative, or zero

In [1]:
```
num = float(input("Enter a number: "))
if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")
```

```
Positive
```

## 3. Find the largest of two numbers

In [2]:
```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
if a > b:
    print("Largest number is:", a)
else:
    print("Largest number is:", b)
```

```
Largest number is: 20.0
```

## 4. Find the largest of three numbers

In [3]:
```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
c = float(input("Enter third number: "))

if a >= b and a >= c:
    print("Largest number is:", a)
elif b >= a and b >= c:
```

```
    print("Largest number is:", b)
else:
    print("Largest number is:", c)
```

```
Largest number is: 30.0
```

# 5. Find the smallest of three numbers

In [4]:
```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
c = float(input("Enter third number: "))

if a <= b and a <= c:
    print("Smallest number is:", a)
elif b <= a and b <= c:
    print("Smallest number is:", b)
else:
    print("Smallest number is:", c)
```

```
Smallest number is: 20.0
```

# 6. Check whether a number is divisible by 5 and 11

In [6]:
```
num = int(input("Enter a number: "))

if num % 5 == 0 and num % 11 == 0:
    print("The number is divisible by both 5 and 11.")
else:
    print("The number is not divisible by both 5 and 11.")
```

```
The number is divisible by both 5 and 11.
```

# 7. Find the factorial of a number

In [24]:
```
num = int(input("Enter a number: "))
factorial = 1
for i in range(1, num + 1):
    factorial *= i
print("Factorial =", factorial)
```

```
Factorial = 120
```

# 8. Find the sum of first N natural numbers

In [9]:
```
n = int(input("Enter N: "))
sum_n = 0
for i in range(1, n + 1):
    sum_n += i

print("Sum =", sum_n)
```

```
Sum = 78
```

# 9. Find the sum of squares of first N natural numbers

In [11]:
```
n = int(input("Enter N: "))
sum_squares = 0
for i in range(1, n + 1):
    sum_squares += i ** 2

print("Sum of squares =", sum_squares)
```

```
Sum of squares = 55
```

# 10. Find the sum of cubes of first N natural numbers

In [12]:
```
n = int(input("Enter N: "))
sum_cubes = 0
for i in range(1, n + 1):
    sum_cubes += i ** 3

print("Sum of cubes =", sum_cubes)
```

```
Sum of cubes = 225
```

# 11. Reverse a number.

In [13]:
```
num = int(input("Enter a number: "))
rev = 0

while num > 0:
    digit = num % 10
    rev = rev * 10 + digit
    num //= 10

print("Reversed number:", rev)
```

```
Reversed number: 987654321
```

# 12. Check whether a number is palindrome.

ज्या संख्येला उलट केल्यानंतरही तीच संख्या मिळते तिला Palindrome Number म्हणतात.

In [14]:
```
num = int(input("Enter a number: "))
original = num
rev = 0

while num > 0:
    digit = num % 10
    rev = rev * 10 + digit
    num //= 10
```

```python
if original == rev:
    print("Palindrome Number")
else:
    print("Not a Palindrome Number")
```

```
Palindrome Number
```

# 13. Check whether a number is Armstrong.

ज्या संख्येतील प्रत्येक अंकाचा घात (power) घेऊन त्यांची बेरीज केल्यावर मूळ संख्या मिळते ती Armstrong Number असते.

उदा.153

$= 1^3 + 5^3 + 3^3$

= 153 ✔️

In [15]:

```python
num = int(input("Enter a number: "))
order = len(str(num))
temp = num
sum_digits = 0

while temp > 0:
    digit = temp % 10
    sum_digits += digit ** order
    temp //= 10

if sum_digits == num:
    print("Armstrong Number")
else:
    print("Not an Armstrong Number")
```

```
Armstrong Number
```

# 14. Check whether a number is Strong Number.

ज्या संख्येतील प्रत्येक अंकाचा factorial घेऊन त्यांची बेरीज मूळ संख्येसारखीच येते ती Strong Number असते.

उदा.145

$= 1! + 4! + 5!$

= 145

In [16]:

```python
import math
num = int(input("Enter a number: "))
temp = num
sum_fact = 0

while temp > 0:
```

```
    digit = temp % 10
    sum_fact += math.factorial(digit)
    temp //= 10

if sum_fact == num:
    print("Strong Number")
else:
    print("Not a Strong Number")
```

```
Strong Number
```

# 15. Check whether a number is Perfect Number.

ज्या संख्येचे सर्व योग्य divisors (स्वतःशिवाय भाग जाणारे अंक) यांची बेरीज मूळ संख्येसारखीच येते ती Perfect Number असते.

उदा.28

= 1 + 2 + 4 + 7 + 14

= 28 ✔

In [18]:
```
num = int(input("Enter a number: "))
sum_divisors = 0

for i in range(1, num):
    if num % i == 0:
        sum_divisors += i

if sum_divisors == num:
    print("Perfect Number")
else:
    print("Not a Perfect Number")
```

```
Perfect Number
```

# 16. Check whether a number is Neon Number.

ज्या संख्येच्या वर्गातील अंकांची बेरीज त्या संख्येइतकीच असते, ती संख्या Neon Number असते. स्पष्टीकरण

9 चा वर्ग = 81

8 + 1 = 9

म्हणून 9 हा Neon Number आहे.

In [19]:
```
num = int(input("Enter a number: "))

square = num * num
sum_digits = 0

while square > 0:
```

```
    sum_digits += square % 10
    square //= 10

if sum_digits == num:
    print("Neon Number")
else:
    print("Not a Neon Number")
```

```
Neon Number
```

# 17. Check whether a number is Spy Number.

ज्या संख्येतील अंकांची बेरीज आणि गुणाकार समान असतो, ती संख्या Spy Number असते.

उदा. 123

1 + 2 + 3 = 6

1 × 2 × 3 = 6

In [20]:
```
num = int(input("Enter a number: "))
sum_digits = 0
product = 1
temp = num

while temp > 0:
    digit = temp % 10
    sum_digits += digit
    product *= digit
    temp //= 10

if sum_digits == product:
    print("Spy Number")
else:
    print("Not a Spy Number")
```

```
Spy Number
```

# 18. Check whether a number is Automorphic Number.

ज्या संख्येच्या वर्गाच्या शेवटचे अंक तीच संख्या दर्शवतात, ती संख्या Automorphic Number असते.

उदा. $25^2 = 625$

In [21]:
```
num = int(input("Enter a number: "))

square = num * num

if str(square).endswith(str(num)):
    print("Automorphic Number")
```

```
else:
    print("Not an Automorphic Number")
```

```
Automorphic Number
```

# 19. Check whether a number is Harshad (Niven) Number.

ज्या संख्येला तिच्या अंकांच्या बेरजेने पूर्ण भाग जातो, ती संख्या Harshad Number असते.

उदा. 18

1 + 8 = 9

18 ÷ 9 = 2

In [22]:
```
num = int(input("Enter a number: "))

sum_digits = 0
temp = num

while temp > 0:
    sum_digits += temp % 10
    temp //= 10

if num % sum_digits == 0:
    print("Harshad Number")
else:
    print("Not a Harshad Number")
```

```
Harshad Number
```

# 20. Find the sum of digits of a number.

संख्येतील सर्व अंकांची बेरीज काढणे.

उदा. 1234

1 + 2 + 3 + 4 = 10

In [23]:
```
num = int(input("Enter a number: "))

sum_digits = 0

while num > 0:
    digit = num % 10
    sum_digits += digit
    num //= 10

print("Sum of digits =", sum_digits)
```

```
Sum of digits = 21
```

# 21. Find the product of digits of a number.

संख्येतील सर्व अंकांचा गुणाकार काढला जातो.

उदा. 123 → 1×2×3 = 6

In [1]:
```
num = int(input("Enter a number: "))

product = 1

while num > 0:
    digit = num % 10
    product *= digit
    num //= 10

print("Product of digits:", product)
```

```
Product of digits: 6
```

# 22. Find the largest digit in a number.

संख्येतला सर्वात मोठा अंक शोधतो.

In [2]:
```
num = int(input("Enter a number: "))

largest = 0

while num > 0:
    digit = num % 10
    if digit > largest:
        largest = digit
    num //= 10

print("Largest digit:", largest)
```

```
Largest digit: 8
```

# 23. Find the smallest digit in a number.

संख्येतला सर्वात लहान अंक शोधतो.

In [3]:
```
num = int(input("Enter a number: "))

smallest = 9

while num > 0:
    digit = num % 10
    if digit < smallest:
        smallest = digit
    num //= 10
```

```
print("Smallest digit:", smallest)
```

```
Smallest digit: 1
```

# 24. Count the number of digits in a number.

संख्येत किती digits आहेत ते मोजतो.

In [4]:
```
num = int(input("Enter a number: "))

count = 0

while num > 0:
    count += 1
    num //= 10

print("Number of digits:", count)
```

```
Number of digits: 5
```

# 25. Find the first and last digit of a number.

Last digit → % 10 ने मिळतो

First digit → शेवटपर्यंत divide करून मिळतो

In [5]:
```
num = int(input("Enter a number: "))

last = num % 10

first = num
while first >= 10:
    first //= 10

print("First digit:", first)
print("Last digit:", last)
```

```
First digit: 1
Last digit: 5
```

# 26. Swap first and last digits of a number.

संख्येतील पहिला आणि शेवटचा अंक बदलतो.

In [6]:
```
num = int(input("Enter a number: "))

last = num % 10
temp = num

digits = len(str(num))
first = temp // (10 ** (digits - 1))
```

```
middle = temp % (10 ** (digits - 1))
middle = middle // 10

swapped = last * (10 ** (digits - 1)) + middle * 10 + first

print("Number after swapping:", swapped)
```

```
Number after swapping: 9420912687
```

# 27. Check whether a number is prime.

ज्या संख्येला 1 आणि स्वतःशिवाय दुसरे कोणतेही भाग जात नाहीत ती Prime Number असते.

उदा. 2, 3, 5, 7, 11 , 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97,

In [9]:
```
num = int(input("Enter a number: "))

if num > 1:
    for i in range(2, num):
        if num % i == 0:
            print("Not a Prime Number")
            break
    else:
        print("Prime Number")
else:
    print("Not a Prime Number")
```

```
Prime Number
```

# 28. Print all prime numbers in a range.

    Prime Number म्हणजे असा संख्या ज्याला 1 आणि स्वतःशिवाय दुसरा कोणताही
    भागाकार जात नाही.

In [1]:
```
start = int(input("Enter starting number: "))
end = int(input("Enter ending number: "))

for num in range(start, end + 1):
    if num > 1:
        is_prime = True

        for i in range(2, int(num ** 0.5) + 1):
            if num % i == 0:
                is_prime = False
                break

        if is_prime:
            print(num, end=" ")
```

```
11 13 17 19 23 29
```

# 29. Find the sum of prime numbers in a range.

रेंजमधील प्रत्येक संख्या Prime आहे का ते तपासून Prime संख्या sum मध्ये जमा करतो.

In [2]:
```
start = int(input("Enter starting number: "))
end = int(input("Enter ending number: "))

total = 0

for num in range(start, end + 1):
    if num > 1:
        is_prime = True

        for i in range(2, int(num ** 0.5) + 1):
            if num % i == 0:
                is_prime = False
                break

        if is_prime:
            total += num

print("Sum of prime numbers =", total)
```

```
Sum of prime numbers = 180
```

# 30. Find GCD (HCF) of two numbers.

GCD (Greatest Common Divisor) म्हणजे दोन्ही संख्यांना पूर्णपणे भाग जाणारी सर्वात मोठी संख्या.

In [3]:
```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

while b != 0:
    a, b = b, a % b

print("GCD =", a)
```

```
GCD = 12
```

# 31. Find LCM of to numbers.

LCM (Least Common Multiple) म्हणजे दोन्ही संख्यांचा सर्वात लहान समान गुणाकार.
LCM = (a × b) / GCD

In [4]:
```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

x, y = a, b

while y != 0:
    x, y = y, x % y
```

```
gcd = x
lcm = (a * b) // gcd

print("LCM =", lcm)
```

```
LCM = 228
```

# 32. Generate Fibonacci series.

> Fibonacci मध्ये प्रत्येक संख्या मागील दोन संख्यांच्या बेरजेइतकी असते.
> सुरुवात 0 आणि 1 पासून होते.

In [7]:
```
n = int(input("Enter number of terms: "))

a = 0
b = 1

for i in range(n):
    print(a, end=" ")
    a, b = b, a + b
```

```
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584
```

# 33. Find Nth Fibonacci number.

> Fibonacci मालिकेतील N व्या स्थानावरील संख्या शोधतो.

In [8]:
```
n = int(input("Enter position: "))

a = 0
b = 1

for i in range(n):
    a, b = b, a + b

print("Nth Fibonacci Number =", a)
```

```
Nth Fibonacci Number = 5
```

# 34. Check whether a number belongs to Fibonacci series.

Fibonacci Series तयार केली जाते. दिलेली संख्या Series मध्ये आढळल्यास ती Fibonacci Number आहे.

In [9]:
```
num = int(input("Enter a number: "))

a = 0
b = 1

while a < num:
    a, b = b, a + b
```

```
if a == num:
    print("Number belongs to Fibonacci Series")
else:
    print("Number does not belong to Fibonacci Series")
```

```
Number belongs to Fibonacci Series
```

## 35. Print multiplication table of a number.

for loop 1 ते 10 पर्यंत चालतो. प्रत्येक वेळी संख्या i ने गुणून table print केली जाते

In [12]:
```
num = int(input("Enter a number: "))

for i in range(1, 11):
    print(f"{num} x {i} = {num * i}")
```

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

# String Programs

## 36. Reverse a string.

[::-1] वापरून string उलटी (reverse) केली जाते.

Reverse झालेली string print केली जाते.

In [14]:
```
text = input("Enter a string: ")

reversed_text = text[::-1]

print("Reversed string:", reversed_text)
```

```
Reversed string: ayhdnas
```

## 37. Check whether a string is palindrome.

alindrome म्हणजे पुढून आणि मागून वाचल्यावर सारखी दिसणारी string.

उदाहरण: madam, radar, level

In [15]:
```
text = input("Enter a string: ")

if text == text[::-1]:
    print("String is a palindrome")
else:
    print("String is not a palindrome")
```

```
String is a palindrome
```

# 38. Count vowels in a string.

Vowels म्हणजे: a, e, i, o, u

प्रत्येक character तपासून vowel असल्यास count वाढवतो.

In [16]:
```
text = input("Enter a string: ")

count = 0

for ch in text:
    if ch.lower() in "aeiou":
        count += 1

print("Number of vowels:", count)
```

```
Number of vowels: 2
```

# 39. Count consonants in a string.

Consonants म्हणजे vowels वगळता उरलेली अक्षरे.

isalpha() वापरून फक्त alphabet तपासले जातात.

In [17]:
```
text = input("Enter a string: ")

count = 0

for ch in text:
    if ch.isalpha() and ch.lower() not in "aeiou":
        count += 1

print("Number of consonants:", count)
```

```
Number of consonants: 3
```

# 40. Count uppercase and lowercase letters.

isupper() → Capital letters मोजते.

islower() → Small letters मोजते.

Space, numbers, special characters मोजले जात नाहीत.

In [1]:
```
text = input("Enter a string: ")

upper_count = 0
lower_count = 0

for ch in text:
    if ch.isupper():
        upper_count += 1
    elif ch.islower():
        lower_count += 1

print("Uppercase letters:", upper_count)
print("Lowercase letters:", lower_count)
```

```
Uppercase letters: 1
Lowercase letters: 4
```

# 41. Count digits in a string.

isdigit() वापरून प्रत्येक character digit आहे का ते तपासले जाते.

Digit आढळल्यास count वाढवला जातो.

In [2]:
```
text = input("Enter a string: ")

digit_count = 0

for ch in text:
    if ch.isdigit():
        digit_count += 1

print("Number of digits:", digit_count)
```

```
Number of digits: 5
```

# 42. Count special characters in a string.

Special Characters: @ # $ % & ! * इ.

isalnum() म्हणजे letter किंवा digit.

जे letter, digit किंवा space नाहीत ते special characters मानले जातात.

In [3]:
```
text = input("Enter a string: ")

special_count = 0

for ch in text:
    if not ch.isalnum() and not ch.isspace():
        special_count += 1
```

```
print("Number of special characters:", special_count)
```

```
Number of special characters: 2
```

# 43. Remove spaces from a string.

replace(" ", "") वापरून सर्व spaces काढून टाकल्या जातात.

In [4]:
```
text = input("Enter a string: ")

result = text.replace(" ", "")

print("String after removing spaces:", result)
```

```
String after removing spaces: sandhyaraghunathmagadum
```

# 44. Remove duplicate characters from a string.

प्रत्येक character तपासला जातो.

तो आधी result मध्ये नसेल तरच add केला जातो.

Duplicate characters काढले जातात.

In [5]:
```
text = input("Enter a string: ")

result = ""

for ch in text:
    if ch not in result:
        result += ch

print("String after removing duplicates:", result)
```

```
String after removing duplicates: progamin
```

# 45. Find frequency of each character.

Dictionary वापरून प्रत्येक character किती वेळा आला आहे ते मोजले जाते.

get(ch, 0) जर character नसला तर 0 परत करते.

Frequency dictionary मध्ये साठवली जाते.

In [6]:
```
text = input("Enter a string: ")

frequency = {}

for ch in text:
    frequency[ch] = frequency.get(ch, 0) + 1
```

```
for key, value in frequency.items():
    print(key, ":", value)
```

```
n : 2
a : 2
y : 1
```

# 46. Find first non-repeated character.

प्रत्येक character किती वेळा आला आहे ते count() ने तपासले जाते.

ज्या character ची frequency 1 आहे तो पहिला non-repeated character असतो.

In [7]:

```
text = input("Enter a string: ")

for ch in text:
    if text.count(ch) == 1:
        print("First non-repeated character:", ch)
        break
else:
    print("No non-repeated character found")
```

```
First non-repeated character: w
```

# 47. Find first repeated character.

Set मध्ये आधी आलेले characters ठेवले जातात.

पुन्हा तोच character आल्यास तो first repeated character असतो.

In [8]:

```
text = input("Enter a string: ")

seen = set()

for ch in text:
    if ch in seen:
        print("First repeated character:", ch)
        break
    seen.add(ch)
else:
    print("No repeated character found")
```

```
First repeated character: r
```

# 48. Check whether two strings are anagrams.

Anagram मध्ये दोन्ही strings मधील अक्षरे समान असतात पण क्रम वेगळा असू शकतो.

sorted() वापरून दोन्ही strings compare केल्या जा

In [9]:
```
str1 = input("Enter first string: ")
str2 = input("Enter second string: ")

if sorted(str1.lower()) == sorted(str2.lower()):
    print("Strings are anagrams")
else:
    print("Strings are not anagrams")
```

```
Strings are anagrams
```

# 49. Replace all vowels with '*'.

प्रत्येक vowel (a, e, i, o, u) च्या जागी * ठेवला जातो.

बाकी characters तसेच ठेवले जातात.

In [10]:
```
text = input("Enter a string: ")

result = ""

for ch in text:
    if ch.lower() in "aeiou":
        result += "*"
    else:
        result += ch

print("Modified string:", result)
```

```
Modified string: s*ndhy*
```

# 50. Convert first character of every word to uppercase

title() function प्रत्येक शब्दाचे पहिले अक्षर Capital करते.

बाकी अक्षरे Small ठेवते.

In [11]:
```
text = input("Enter a sentence: ")

result = text.title()

print("Converted string:", result)
```

```
Converted string: Welcome To Python Programming
```

# 51. Find longest word in a sentence.

split() वापरून sentence मधील सर्व शब्द वेगळे केले जातात.

max(words, key=len) सर्वात मोठा शब्द शोधते.

In [12]:
```
sentence = input("Enter a sentence: ")

words = sentence.split()

longest_word = max(words, key=len)

print("Longest word:", longest_word)
```

```
Longest word: raghunath
```

# 52. Find shortest word in a sentence.

min(words, key=len) सर्वात लहान शब्द शोधते.

In [15]:
```
sentence = input("Enter a sentence: ")

words = sentence.split()

shortest_word = min(words, key=len)

print("Shortest word:", shortest_word)
```

```
Shortest word: is
```

# 53. Count number of words in a sentence

split() नंतर len() वापरून शब्दांची संख्या मोजली जाते.

In [16]:
```
sentence = input("Enter a sentence: ")

words = sentence.split()

print("Number of words:", len(words))
```

```
Number of words: 6
```

# 54. Reverse each word in a sentence.

प्रत्येक शब्दासाठी [::-1] वापरून तो उलटा केला जातो.

शब्दांचा क्रम बदलत नाही.

In [19]:
```
sentence = input("Enter a sentence: ")

words = sentence.split()

for word in words:
    print(word[::-1], end=" ")
```

```
ayhdnas dna ahsin
```

# 55. Reverse the entire sentence.

संपूर्ण string वर [::-1] वापरले जाते.

त्यामुळे पूर्ण sentence उलटी होते.

In [18]:
```
sentence = input("Enter a sentence: ")

reversed_sentence = sentence[::-1]

print("Reversed sentence:", reversed_sentence)
```

```
Reversed sentence: ahsin dna ayhdnas
```

# 56. Print star triangle pattern.

प्रत्येक row मध्ये stars ची संख्या वाढत जाते.

1 पासून rows पर्यंत loop चालतो.

In [1]:
```
rows = int(input("Enter number of rows: "))

for i in range(1, rows + 1):
    print("*" * i)
```

```
*
**
***
****
*****
******
```

# 57. Print inverted triangle pattern.

Stars ची संख्या प्रत्येक row मध्ये कमी होत जाते.

In [2]:
```
rows = int(input("Enter number of rows: "))

for i in range(rows, 0, -1):
    print("*" * i)
```

```
*****
****
***
**
*
```

# 58. Print pyramid pattern.

Spaces आणि Stars वापरून Pyramid तयार केली जाते.

प्रत्येक row मध्ये stars ची संख्या 2 ने वाढते.

In [3]:
```
rows = int(input("Enter number of rows: "))

for i in range(rows):
    print(" " * (rows - i - 1) + "*" * (2 * i + 1))
```

```
    *
   ***
  *****
 *******
*********
```

# 59. Print number triangle pattern.

प्रत्येक row मध्ये 1 पासून त्या row क्रमांकापर्यंत संख्या print होतात.

In [4]:
```
rows = int(input("Enter number of rows: "))

for i in range(1, rows + 1):
    for j in range(1, i + 1):
        print(j, end=" ")
    print()
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

# 60. Print Floyd's triangle.

Numbers सतत वाढत जातात.

प्रत्येक row मध्ये row number इतक्या संख्या print होतात.

In [5]:
```
rows = int(input("Enter number of rows: "))

num = 1

for i in range(1, rows + 1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

# 61. Print Pascal's triangle.

Pascal Triangle मध्ये प्रत्येक संख्या वरच्या दोन संख्यांच्या बेरजेइतकी असते.

In [6]:
```
rows = int(input("Enter number of rows: "))

for i in range(rows):
    num = 1

    print(" " * (rows - i), end="")

    for j in range(i + 1):
        print(num, end=" ")
        num = num * (i - j) // (j + 1)

    print()
```

```
     1
    1 1
   1 2 1
  1 3 3 1
 1 4 6 4 1
1 5 10 10 5 1
```

# 62. Print diamond pattern.

Pyramid + Inverted Pyramid = Diamond Pattern

In [7]:
```
rows = int(input("Enter number of rows: "))

for i in range(rows):
    print(" " * (rows - i - 1) + "*" * (2 * i + 1))

for i in range(rows - 2, -1, -1):
    print(" " * (rows - i - 1) + "*" * (2 * i + 1))
```

```
   *
  ***
 *****
*******
 *****
  ***
   *
```

# 63. Print hollow square pattern.

Border वर stars print होतात.

आतमध्ये spaces print होतात.

In [8]:
```
size = int(input("Enter size: "))

for i in range(size):
    for j in range(size):
        if i == 0 or i == size - 1 or j == 0 or j == size - 1:
            print("*", end=" ")
        else:
            print(" ", end=" ")
    print()
```

```
* * * * * *
*         *
*         *
*         *
*         *
* * * * * *
```

# 64. Print hollow pyramid pattern.

फक्त boundary वर stars print होतात.

आतमध्ये spaces राहतात.

In [9]:
```
rows = int(input("Enter number of rows: "))

for i in range(rows):
    for j in range(2 * rows - 1):

        if j == rows - i - 1 or j == rows + i - 1:
            print("*", end="")

        elif i == rows - 1 and rows - i - 1 <= j <= rows + i - 1:
            print("*", end="")

        else:
            print(" ", end="")

    print()
```

```
      *
     * *
    *   *
   *     *
  *       *
 *         *
*************
```

# 65. Print alphabet pattern.

chr(65) = A

chr(66) = B

chr(67) = C

प्रत्येक row मध्ये alphabet वाढत जातात.

In [10]:
```
rows = int(input("Enter number of rows: "))

for i in range(rows):
    for j in range(i + 1):
        print(chr(65 + j), end=" ")
    print()
```

```
A
A B
A B C
A B C D
A B C D E
A B C D E F
```

# List Programs

## 66. Find largest element in a list.

पहिला element largest मानला जातो.

प्रत्येक element शी तुलना करून मोठा element शोधला जातो.

In [11]:
```
numbers = [10, 45, 23, 89, 12]

largest = numbers[0]

for num in numbers:
    if num > largest:
        largest = num

print("Largest element:", largest)
```

```
Largest element: 89
```

## 67. Find smallest element in a list.

पहिला element smallest मानला जातो.

प्रत्येक element शी तुलना करून सर्वात लहान element शोधला जातो.

In [12]:
```
numbers = [10, 45, 23, 89, 12]

smallest = numbers[0]

for num in numbers:
    if num < smallest:
```

```
        smallest = num

print("Smallest element:", smallest)
```

```
Smallest element: 10
```

# 68. Find second largest element.

Duplicate values काढल्या जातात.

List sort करून शेवटून दुसरा element घेतला जातो.

In [13]:
```
numbers = [10, 45, 23, 89, 12]

unique_numbers = list(set(numbers))
unique_numbers.sort()

print("Second largest element:", unique_numbers[-2])
```

```
Second largest element: 45
```

# 69. Find second smallest element.

Sort केल्यानंतर index 1 वरील element second smallest असतो.

In [14]:
```
numbers = [10, 45, 23, 89, 12]

unique_numbers = list(set(numbers))
unique_numbers.sort()

print("Second smallest element:", unique_numbers[1])
```

```
Second smallest element: 12
```

# 70. Remove duplicates from a list.

Element आधीपासून list मध्ये नसेल तरच add केला जातो.

In [15]:
```
numbers = [10, 20, 30, 20, 40, 10, 50]

unique_list = []

for num in numbers:
    if num not in unique_list:
        unique_list.append(num)

print("List after removing duplicates:", unique_list)
```

```
List after removing duplicates: [10, 20, 30, 40, 50]
```

# 71. Find common elements between two lists.

दोन्ही list मध्ये असलेले elements शोधले जातात.

In [16]:
```
list1 = [10, 20, 30, 40]
list2 = [30, 40, 50, 60]

common = []

for num in list1:
    if num in list2:
        common.append(num)

print("Common elements:", common)
```

```
Common elements: [30, 40]
```

# 72. Merge two lists.

operator plus + वापरून दोन lists एकत्र केल्या जातात.

In [17]:
```
list1 = [10, 20, 30]
list2 = [40, 50, 60]

merged_list = list1 + list2

print("Merged list:", merged_list)
```

```
Merged list: [10, 20, 30, 40, 50, 60]
```

# 73. Sort list without using sort().

Nested loop वापरून elements swap केले जातात.

Bubble Sort सारखी logic वापरली आहे.

In [18]:
```
numbers = [45, 12, 89, 23, 10]

for i in range(len(numbers)):
    for j in range(i + 1, len(numbers)):
        if numbers[i] > numbers[j]:
            numbers[i], numbers[j] = numbers[j], numbers[i]

print("Sorted list:", numbers)
```

```
Sorted list: [10, 12, 23, 45, 89]
```

# 74. Reverse a list without using reverse().

शेवटच्या index पासून सुरुवात करून नवीन list तयार केली जाते.

In [19]:
```
numbers = [10, 20, 30, 40, 50]
```

```
reversed_list = []

for i in range(len(numbers) - 1, -1, -1):
    reversed_list.append(numbers[i])

print("Reversed list:", reversed_list)
```

```
Reversed list: [50, 40, 30, 20, 10]
```

# 75. Find sum of all elements in a list.

प्रत्येक element total मध्ये add केला जातो.

In [20]:
```
numbers = [10, 20, 30, 40, 50]

total = 0

for num in numbers:
    total += num

print("Sum =", total)
```

```
Sum = 150
```

# 76. Find even numbers in a list.

num % 2 == 0 असल्यास संख्या even असते.

In [21]:
```
numbers = [10, 15, 20, 25, 30, 35]

print("Even numbers:")

for num in numbers:
    if num % 2 == 0:
        print(num, end=" ")
```

```
Even numbers:
10 20 30
```

# 77. Find odd numbers in a list.

num % 2 != 0 असल्यास संख्या odd असते.

In [22]:
```
numbers = [10, 15, 20, 25, 30, 35]

print("Odd numbers:")

for num in numbers:
    if num % 2 != 0:
        print(num, end=" ")
```

```
Odd numbers:
15 25 35
```

# 78. Count positive and negative numbers in a list.

num > 0 → Positive Count

num < 0 → Negative Count

In [23]:
```
numbers = [10, -5, 20, -15, 30, -2]

positive = 0
negative = 0

for num in numbers:
    if num > 0:
        positive += 1
    elif num < 0:
        negative += 1

print("Positive numbers:", positive)
print("Negative numbers:", negative)
```

```
Positive numbers: 3
Negative numbers: 3
```

# Dictionary Programs

# 79. Count frequency of words in a sentence.

split() वापरून sentence मधील शब्द वेगळे केले जातात.

Dictionary मध्ये प्रत्येक शब्द किती वेळा आला आहे ते मोजले जाते.

In [24]:
```
sentence = input("Enter a sentence: ")

words = sentence.split()

frequency = {}

for word in words:
    frequency[word] = frequency.get(word, 0) + 1

print("Word Frequencies:")
for key, value in frequency.items():
    print(key, ":", value)
```

```
Word Frequencies:
python : 2
is : 2
easy : 1
and : 1
powerful : 1
```

# 80. Find key with maximum value.

data.get च्या आधारावर max() सर्वात मोठी value असलेली key शोधते.

In [25]:
```
data = {
    "A": 100,
    "B": 250,
    "C": 150,
    "D": 300
}

max_key = max(data, key=data.get)

print("Key with maximum value:", max_key)
print("Maximum value:", data[max_key])
```

```
Key with maximum value: D
Maximum value: 300
```

# 81. Merge two dictionaries.

** operator वापरून दोन dictionaries एकत्र केल्या जातात.

In [26]:
```
dict1 = {"A": 10, "B": 20}
dict2 = {"C": 30, "D": 40}

merged_dict = {**dict1, **dict2}

print("Merged Dictionary:")
print(merged_dict)
```

```
Merged Dictionary:
{'A': 10, 'B': 20, 'C': 30, 'D': 40}
```

# 82. Sort dictionary by keys.

sorted() dictionary च्या keys alphabetically sort करते.

In [27]:
```
data = {
    "D": 40,
    "A": 10,
    "C": 30,
    "B": 20
}
```

```
sorted_dict = dict(sorted(data.items()))

print("Dictionary Sorted by Keys:")
print(sorted_dict)
```

```
Dictionary Sorted by Keys:
{'A': 10, 'B': 20, 'C': 30, 'D': 40}
```

# 83. Sort dictionary by values.

item[1] म्हणजे dictionary ची value.

Values च्या ascending order नुसार sorting होते.

In [28]:
```
data = {
    "A": 40,
    "B": 10,
    "C": 30,
    "D": 20
}

sorted_dict = dict(sorted(data.items(), key=lambda item: item[1]))

print("Dictionary Sorted by Values:")
print(sorted_dict)
```

```
Dictionary Sorted by Values:
{'B': 10, 'D': 20, 'C': 30, 'A': 40}
```

# 84. Invert a dictionary (swap keys and values).

Original dictionary मधील key आणि value ची जागा बदलली जाते.

New dictionary मध्ये value key बनते आणि key value बनते.

In [29]:
```
data = {
    "A": 10,
    "B": 20,
    "C": 30
}

inverted_dict = {}

for key, value in data.items():
    inverted_dict[value] = key

print("Inverted Dictionary:")
print(inverted_dict)
```

```
Inverted Dictionary:
{10: 'A', 20: 'B', 30: 'C'}
```

# Type Casting & Data Types

## 85. Convert string to integer.

int() function वापरून string ला integer मध्ये convert केले जाते.

In [30]:
```
text = "123"

number = int(text)

print(number)
print(type(number))
```

```
123
<class 'int'>
```

## 86. Convert integer to float.

float() वापरून integer ला float मध्ये convert केले जाते.

In [31]:
```
num = 100

result = float(num)

print(result)
print(type(result))
```

```
100.0
<class 'float'>
```

## 87. Convert float to integer.

int() दशांश भाग काढून टाकते.

45.89 → 45

In [32]:
```
num = 45.89

result = int(num)

print(result)
print(type(result))
```

```
45
<class 'int'>
```

## 88. Convert list to tuple.

tuple() वापरून list चे tuple मध्ये रूपांतर होते.

In [33]:
```
my_list = [10, 20, 30, 40]

my_tuple = tuple(my_list)

print(my_tuple)
print(type(my_tuple))
```

```
(10, 20, 30, 40)
<class 'tuple'>
```

# 89. Convert tuple to list.

list() वापरून tuple चे list मध्ये रूपांतर होते.

In [34]:
```
my_tuple = (10, 20, 30, 40)

my_list = list(my_tuple)

print(my_list)
print(type(my_list))
```

```
[10, 20, 30, 40]
<class 'list'>
```

# 90. Convert string to list.

list() string मधील प्रत्येक character ला वेगळा element बनवते.

In [35]:
```
text = "Python"

result = list(text)

print(result)
```

```
['P', 'y', 't', 'h', 'o', 'n']
```

# 91. Convert list to string.

join() वापरून list मधील elements एकत्र करून string तयार केली जाते.

In [36]:
```
my_list = ['P', 'y', 't', 'h', 'o', 'n']

result = "".join(my_list)

print(result)
```

```
Python
```

# 92. Find data type of user input.

input() ने घेतलेले value नेहमी String असते.

In [38]:
```
value = input("Enter any value: ")

print("Data Type:", type(value))
```

```
Data Type: <class 'str'>
```

## 93. Check whether a variable is int, float, or string.

isinstance() वापरून variable चा data type तपासला जातो.

In [39]:
```
value = 45.5

if isinstance(value, int):
    print("Integer")
elif isinstance(value, float):
    print("Float")
elif isinstance(value, str):
    print("String")
else:
    print("Other Data Type")
```

```
Float
```

## 94. Demonstrate mutable and immutable data types.

List मधील values बदलता येतात.

म्हणून List हा Mutable Data Type आहे.

In [40]:
```
numbers = [10, 20, 30]

numbers[0] = 100

print(numbers)
```

```
[100, 20, 30]
```

# If-Else Based Interview Questions

## 95. Grade calculator using marks.

Marks च्या range नुसार Grade दिला जातो.

In [1]:
```
marks = float(input("Enter marks: "))
```

```python
if marks >= 90:
    print("Grade A")
elif marks >= 80:
    print("Grade B")
elif marks >= 70:
    print("Grade C")
elif marks >= 60:
    print("Grade D")
else:
    print("Fail")
```

```
Fail
```

# 96. ATM withdrawal validation.

Amount balance पेक्षा कमी किंवा समान असल्यास withdrawal होते.

In [2]:
```python
balance = 10000

amount = float(input("Enter withdrawal amount: "))

if amount <= balance:
    print("Withdrawal Successful")
    print("Remaining Balance:", balance - amount)
else:
    print("Insufficient Balance")
```

```
Withdrawal Successful
Remaining Balance: 7000.0
```

# 97. Electricity bill calculation.

In [3]:
```python
units = int(input("Enter units consumed: "))

if units <= 100:
    bill = units * 5
elif units <= 200:
    bill = (100 * 5) + ((units - 100) * 7)
else:
    bill = (100 * 5) + (100 * 7) + ((units - 200) * 10)

print("Electricity Bill =", bill)
```

```
Electricity Bill = 2200
```

# 98. Income tax calculation.

Income slab नुसार tax rate बदलतो.

In [4]:
```python
income = float(input("Enter annual income: "))

if income <= 250000:
```

```
    tax = 0
elif income <= 500000:
    tax = income * 0.05
elif income <= 1000000:
    tax = income * 0.20
else:
    tax = income * 0.30

print("Income Tax =", tax)
```

```
Income Tax = 120000.0
```

# 99. Age category checker (Child, Teen, Adult, Senior).

In [5]:

```
age = int(input("Enter age: "))

if age < 13:
    print("Child")
elif age < 20:
    print("Teen")
elif age < 60:
    print("Adult")
else:
    print("Senior Citizen")
```

```
Adult
```

# 100. Login system using username and password.

In [6]:

```
username = input("Enter username: ")
password = input("Enter password: ")

if username == "admin" and password == "1234":
    print("Login Successful")
else:
    print("Invalid Username or Password")
```

```
Login Successful
```

# 101. Reverse a number.

In [8]:

```
num = int(input("Enter a number: "))

reverse = 0

while num > 0:
    digit = num % 10
    reverse = reverse * 10 + digit
    num //= 10

print("Reversed Number =", reverse)
```

```
Reversed Number = 12345
```

# 102. Check whether a number is a palindrome.

In [9]:
```
num = int(input("Enter a number: "))

original = num
reverse = 0

while num > 0:
    digit = num % 10
    reverse = reverse * 10 + digit
    num //= 10

if original == reverse:
    print("Palindrome Number")
else:
    print("Not a Palindrome Number")
```

```
Palindrome Number
```

# 103. Check whether a number is an Armstrong number.

$153 = 1^3 + 5^3 + 3^3 = 153$

In [10]:
```
num = int(input("Enter a number: "))

original = num
digits = len(str(num))
total = 0

while num > 0:
    digit = num % 10
    total += digit ** digits
    num //= 10

if total == original:
    print("Armstrong Number")
else:
    print("Not an Armstrong Number")
```

```
Armstrong Number
```

# 104. Find the sum of digits of a number.

In [11]:
```
num = int(input("Enter a number: "))

total = 0

while num > 0:
    total += num % 10
```

```
    num //= 10

print("Sum of Digits =", total)
```

Sum of Digits = 15

## 105. Find the largest digit in a number.

In [12]:
```
num = int(input("Enter a number: "))

largest = 0

while num > 0:
    digit = num % 10

    if digit > largest:
        largest = digit

    num //= 10

print("Largest Digit =", largest)
```

Largest Digit = 9

## 106. Print Fibonacci series.

In [13]:
```
n = int(input("Enter number of terms: "))

a = 0
b = 1

for i in range(n):
    print(a, end=" ")
    a, b = b, a + b
```

0 1 1 2 3

## 107. Find prime numbers in a given range.

In [14]:
```
start = int(input("Enter start number: "))
end = int(input("Enter end number: "))

for num in range(start, end + 1):

    if num > 1:
        is_prime = True

        for i in range(2, int(num ** 0.5) + 1):
            if num % i == 0:
                is_prime = False
                break
```

```
        if is_prime:
            print(num, end=" ")
```

17 19 23 29 31

## 108. Count vowels, consonants, digits, and special characters in a string.

In [15]:
```
text = input("Enter a string: ")

vowels = consonants = digits = special = 0

for ch in text:
    if ch.lower() in "aeiou":
        vowels += 1
    elif ch.isalpha():
        consonants += 1
    elif ch.isdigit():
        digits += 1
    elif not ch.isspace():
        special += 1

print("Vowels =", vowels)
print("Consonants =", consonants)
print("Digits =", digits)
print("Special Characters =", special)
```

```
Vowels = 2
Consonants = 3
Digits = 3
Special Characters = 1
```

## 109. Remove duplicate characters from a string.

In [16]:
```
text = input("Enter a string: ")

result = ""

for ch in text:
    if ch not in result:
        result += ch

print("Result =", result)
```

Result = progamin

## 110. Find the frequency of each character in a string.

In [18]:
```
text = input("Enter a string: ")

frequency = {}
```

```python
for ch in text:
    frequency[ch] = frequency.get(ch, 0) + 1

for key, value in frequency.items():
    print(key, ":", value)
```

```
b : 1
a : 3
n : 2
```

In [ ]: